

The Applied Biosystems 4700 Proteomics Analyzer—High Productivity to Meet the Proteomics Challenge

Purpose

The Applied Biosystems 4700 Proteomics Analyzer is uniquely designed to deliver on the promise of proteomics by providing high-resolution spectral information while maximizing sample throughput. Experiments on the Applied Biosystems 4700 Proteomics Analyzer accomplish in days what previously took months to complete.

Overview

Proteomics encompasses the identification and differential expression of proteins, protein-protein interactions, and structural and functional protein analysis. General estimates from genomic studies suggest that the 30,000 genes of human DNA could transcribe to more than 1,000,000 proteins. Although proteomics holds enormous promise for advancements in life science, protein identification and detailed characterization present complex and time-consuming problems. This level of characterization and analysis presents new challenges in deciphering the complexity and new requirements for sample throughput.

Key Features

- Combines matrix-assisted laser desorption ionization (MALDI) for ease-of-use and high throughput with “true” MS/MS specificity for continuous automated protein identification and characterization
- Patented TOF/TOF™ Optics provide reproducible high-energy fragmentation, generating information-rich fragmentation patterns for accurate protein identification and enhanced characterization

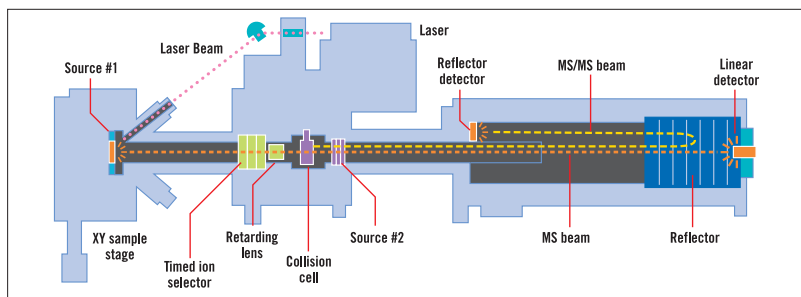


Figure 1. The TOF/TOF™ Optics of the Applied Biosystems 4700 Proteomics Analyzer feature a completely new design in reflectron ion focusing for MS and MS/MS acquisitions.

- Compatible with existing proteomic workflows and enables new, innovative workflows such as those based on the proprietary ICAT™ reagent technology for new levels of productivity in proteomics
- Excellent long term instrument stability for automated high productivity analysis
- High-sensitivity analysis for low copy proteins

Experimental Conditions

In a productivity study, three instruments were set up to run continuously in 24-hour-per-day operation over a five day period. In this experiment, 0.5 μ L of a 250 fmol/ μ L β -galactosidase sample was spotted to each well of a 192-well MALDI sample plate. Twenty-four MALDI plates were inserted into the 4700 Proteomics Analyzer autoloader. In each sample location, 400 shots for MS acquisition were taken, and ten precursors for MS/MS were selected and acquired. MS acquisition for the entire plate took 6.4 minutes with 76,800 laser shots taken per plate. MS/MS acquisition of the ten precursors at 2,000 shots each took 5.33 hours per plate with 3.84 million laser

shots taken. More than 150,000 spectra were acquired for the three instruments over this five-day period.

Results and Discussion

Long-term stability in instrument operation and performance is crucial to meet the high productivity requirements for protein identification and characterization. In this experiment, results from three Applied Biosystems 4700 Proteomics Analyzers that were run continuously over a five day period were compiled and compared. Results were quantified by tracking signal-to-noise and resolution of the 1396 Da precursor ion and the 706 Da fragment ion in every tenth spectra. The performance stability of the 4700 Proteomics Analyzer is demonstrated in Table 1.

The Applied Biosystems 4700 Proteomics Analyzer is the first instrument that uses TOF/TOF Optics for automated MS and MS/MS acquisition and analysis. TOF/TOF Optics are the patented technology that provide the high level of performance of the Applied Biosystems 4700 Proteomics Analyzer. Completely new in design, this radical innovation in the MALDI technique is built on the traditions of the Applied Biosystems

	Instrument 1 MS	Instrument 1 MS/MS	Instrument 2 MS	Instrument 2 MS/MS	Instrument 3 MS	Instrument 3 MS/MS
# Spectra	3,760	3,830	4,130	6,151	4,013	4,514
Resolution: Ave./RSD	12,822 +/- 15%	4,484 +/- 11%	13,278 +/- 14%	4,079 +/- 8%	12,182 +/- 8%	3,980 +/- 9%
S/N: Ave./RSD	250 +/- 27%	177 +/- 24%	191 +/- 20%	106 +/- 34%	132 +/- 22%	154 +/- 33%

Table 1. Summary of results from the automated acquisition of a standard digest of β -galactosidase run continuously on three instruments for five days demonstrates the long-term instrument stability required for high productivity proteomics. Every tenth spectrum was included in the results calculation.

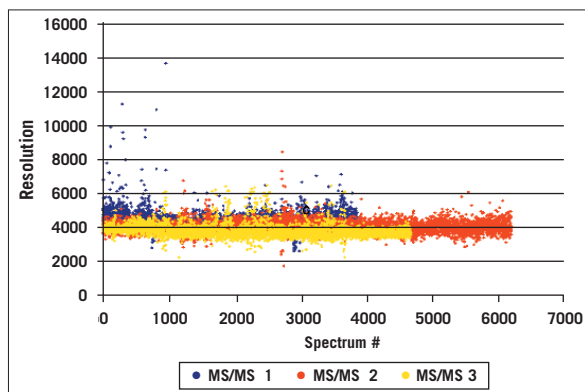


Figure 2. MS/MS resolution plot shows excellent stability of three Applied Biosystems 4700 Proteomics Analyzers over five days of continuous operation.

MALDI-TOF mass spectrometers. High-speed acquisition rates and controllable ion fragmentation using this technology permit unambiguous structural determination not available from other MALDI-TOF technologies.

In addition, the Applied Biosystems 4700 Proteomics Analyzer has the ability to reproducibly modify collision conditions. The degree of fragmentation and the structural information obtained are related to adjustments in laser intensity, collision gas pressure, and collision energy. Valuable information about amino acid composition can be obtained by collating the MS/MS data of varying collision conditions, thus providing confident peptide sequence interpretation and identification of multiple proteins from complex mixtures.

The Applied Biosystems 4700 Proteomics Analyzer also includes a 200 Hz laser and

acquisition system. At 200 Hz, more than 10,000 spectra can be acquired over a 24-hour period. Traditional MALDI-TOF instruments acquire at rates of less than 10 Hz, so comparable numbers of acquisitions would take weeks rather than days to complete. In this experiment, an unprecedented amount of high-quality mass spectral data was acquired in an incredibly short period of time, demonstrating that the Applied Biosystems 4700 Proteomics Analyzer has

unique ability to change the scope and depth of analysis in the field of proteomics.

Conclusions

The Applied Biosystems 4700 Proteomics Analyzer with TOF/TOF™ Optics is the first analyzer designed to meet the scale and scope of the proteomics challenge. The fully automated platform permits the acquisition of reproducible, information-rich, high-quality MS and MS/MS spectra on a reliable high-throughput, easy-to-use MALDI platform. The 4700 Proteomics Analyzer makes it possible for you to increase the speed of sample analysis, expanding both the range of possible projects and depth of biologically significant information generated from complex proteomic experiments. Ultimately, this fully automated platform facilitates a better understanding of biology and the biology of disease.

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Printed in the USA, 2/2003, LD
Publication 115PB03-01